



Towards a process selection method for embedded analytics

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Abstract

Driven by technological progress, business analytics is gaining momentum while paving the path for next-generation business process management. Especially, embedded real-time analytics offers new opportunities for business process intelligence and value creation. However, there are several obstacles that organizations face in their adoption process. A key challenge is to identify business processes that are suitable for embedded analytics and hold relevant value potential. Our research addresses this need by introducing an exploratory BPM method, namely a process selection method. Applying action design research and situational method engineering, we iteratively built, used, evaluated, and refined the theory-ingrained method artifact. The method provides organizations with guidance in selecting operational business processes, for which a reengineering project should be initiated.

Keywords Process selection method · Embedded analytics · Intelligent business processes · Explorative BPM · Action design research

1 Introduction

Fast-paced technological advancements are unlocking new opportunities for value creation, while driving the digital transformation of the workplace (vom Brocke et al. 2018). These developments come along with the rise of business analytics (BA) across various industries and corporate functions (Ng et al. 2021; Ransbotham et al. 2018). BA allows automating and augmenting complex tasks or decision-making using mathematical-statistical models and algorithms (Mortenson et al. 2015). While BA was initially applied to decision problems in strategic management, the reach has gradually expanded to the operational and tactical levels (Sharma et al.

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2014). Within the emerging hyperautomation phenomenon, there is an increasing momentum for embedded analytics applications empowered by enterprise cloud computing (Tarafdar et al. 2019). In general, the term ‘embedded analytics’ is described as a “[...] digital workplace capability where data analysis occurs within a user’s natural workflow, without the need to toggle to another application” (Gartner 2023b). Especially, real-time analytics applications embedded into process models enable to “inject intelligence” into operational business processes, driving operational efficiency through process automation and optimization (Coombs et al. 2020; Enholm et al. 2021). In this article, we refer to embedded analytics as a synonym for this special form of real-time analytics applications that are embedded into operational process flows. This new class of intelligent business process systems employ novel technologies such as cloud computing, artificial intelligence (AI), or machine learning (ML) algorithms—and outperform traditional business processes in terms of adaptability, flexibility, and transparency (Buxmann et al. 2021; Dwivedi et al. 2021; Lepenioti et al. 2020; vom Brocke et al. 2018). Optimizing internal business operations is projected to be a key value source in the emerging AI era (Brock & von Wangenheim 2019; Davenport & Ronanki 2018; Microstrategy 2020). Hence, embedded analytics is an important lever for transforming traditional operating models (Iansiti & Lakhani 2020).

However, the rise of embedded analytics is still in its early days and adoption is often limited (Brock & von Wangenheim 2019; Microstrategy 2020). Organizations face numerous challenges, which slow down adoption and hamper value creation (Davenport & Ronanki 2018; Duan et al. 2019; Microstrategy 2020; Vidgen et al. 2017). For instance, a critical factor is the lack of required skills and knowledge (Brock & von Wangenheim 2019). Recent research also suggests that there are remarkable differences in the business impact achieved after implementation, while the likelihood for significant financial benefits was found to rapidly increase with the extent of business process changes (Ransbotham et al. 2020). Thus, organizations particularly need support at the very beginning of their transformation journey towards intelligent business processes and operations.

Our research addresses this gap pursuing the following research question: “*How to identify operational business processes that are suitable for embedded analytics and how to estimate their potential value?*” In this paper, we propose a method artifact supporting the identification and selection of processes for a process reengineering/innovation project. Accordingly, it is a process selection method in shape of a decision support tool. At the core of our exploratory method is a process assessment model. It includes a novel construct for measuring the ‘task-analytics-fit’, representing an important innovation of our research. Applying action design research (ADR) and situational method engineering (SME) methodology, we built and evaluated the method artifact in two design cycle within an embedded in-depth case study. The initial method artifact was created by selecting and composing existing method fragments or chunks and further evolved across two design cycles. The method artifact was applied and evaluated in real-world interventions with two country organizations of our case company. Thus, we contribute with our process selection method to the prescriptive knowledge base on exploratory BPM and embedded analytics, while addressing a important real-world problem. The remainder of this paper is

structured as follows: In Sect. 2, we present relevant research related to explorative BPM methods and embedded analytics. In Sect. 3, the research methodology and our two-cycle ADR process are outlined. The engineered method artifact including the process assessment model are formalized and explained in Sect. 4. In Sect. 5, we present the application of the process selection method within our case study context. In Sect. 6, we discuss the findings of our research as well as its implications and limitations. Furthermore, we highlight future research directions for our method artifact. In Sect. 7, we conclude by briefly summarizing our research.

2 Related research

2.1 Embedded analytics for enhancing business operations

Business analytics (BA) integrates several academic disciplines that deal with technology, quantitative methods, and decision-making/behavioral theory (Mortenson et al. 2015; Power et al. 2018). Analytics applications or algorithms can be considered as a mediator that translates human input into a machine-readable language. They enable machines to “learn” human knowledge, perform complex information processing and analysis tasks, and replicate human reasoning or decision-making (Brynjolfsson & McAfee 2017; Wirtz et al. 2018). Four types of BA applications are distinguished based on their problem types and time perspective: descriptive, diagnostic, predictive, and prescriptive analytics (Banerjee et al. 2013; Delen & Demirkan 2013). The BA type and used quantitative methods depend on the specific use case and its underlying business logic. Furthermore, it is assumed that complexity and value contribution increase along these four BA stages. *Descriptive analytics* and *diagnostic analytics* are known as *business intelligence* (BI); they are backward-looking in nature and are applied for data/information processing and analytical tasks. *Advanced analytics* refers to *predictive analytics* and *prescriptive analytics*, which generate probabilistic outcomes for future-oriented problems. In last two decades, the growing data volume and rapid technological advances have substantially extended the scope and performance of BA applications, fostering the rise of BA in a wide range of industries and business functions (Chen et al. 2012; Ng et al. 2021; Ransbotham et al. 2018; Wirtz et al. 2018). In particular, the use and relevance of BA have increased in corporate management and business operations (Agrawal et al. 2017; Davenport & Ronanki 2018). Advanced analytics combined with AI technologies, ML algorithms, and streaming computational environments hold significant value potential (Dwivedi et al. 2021); real-time analytics embedded into business process environments are used for automating and augmenting human work or guiding operational workflows (Bichler et al. 2017; Grover et al. 2018). BI has been used in BPM software as integral part of several features or tools. Moreover, analytics applications constitute a key element of recently emerged process mining software that support process discovery and analysis. These application areas adopt an external, “outside-in” perspective serving the analysis and improvement of business processes. However, there is a noticeable shift towards the “inside-out” perspective. In the context of the hyperautomation phenomenon, interest in and relevance

of intelligent business processes leveraging embedded real-time analytics have substantially enhanced. *Hyperautomation* describes the large-scale automation and optimization of business processes leveraging several technologies and tools such as RPA, ML, AI, or BPM suites (Gartner 2023a). The scope of hyperautomation as well as the classification of individual use cases can be defined along the *automation continuum* (Lacity & Willcocks 2016, 2021). Over the last few years, there has been a shift in focus from simple repetitive tasks and RPA to more difficult activities using advanced technologies; this trend is also referred to as intelligent automation (Coombs et al. 2020; Lacity & Willcocks 2021). Intelligent process systems driven by embedded real-time analytics represent a new class of information systems that are characterized by adaptability and flexibility (Buxmann et al. 2021; Ng et al. 2021; van der Aalst et al. 2018). The optimization of internal business operations is expected to provide significant business value, but at the same time it will fundamentally change the way companies operate and compete (Davenport & Ronanki 2018; Iansiti & Lakhani 2020). Humans are not just replaced by machines in various tasks, but instead new work systems with modified forms of human–machine collaboration will emerge (vom Brocke et al. 2018).

2.2 Explorative BPM methods

Business Process Management (BPM) is a holistic management discipline that has traditionally focused on improving business processes and on building needed capabilities (Dumas et al. 2018; vom Brocke et al. 2014). Accordingly, it is critical for effective and efficient organizational performance. Moreover, BPM has a key role for realizing business value from technology through process redesign and innovation (Schmiedel and vom Brocke 2015; Schryen 2013). The *capability perspective* focuses on the prerequisites for successful and meaningful BPM use in organizations. There are 36 capability areas that span across the six elements of BPM capability, namely *strategic alignment, governance, methods, information technology, people, and culture* (Rosemann & vom Brocke, 2015). These elements are critical to success and at the same time highly context-sensitive, implying that BPM initiatives and projects can fail if they do not adequately address the specific organizational and situational context (vom Brocke et al. 2016). The BPM context is determined based on various context factors that are grouped in four context dimensions, i.e., *goal, process, organization, and environment* (vom Brocke et al. 2016). While the traditional goal of improving current business processes reflects a problem-driven approach (exploitation), explorative BPM follows an opportunity-driven approach that might stimulate process redesign or innovation (Benner & Tushman 2003; Grisold et al. 2019). Accordingly, exploitation and exploration have been considered as two distinct goals of BPM; the interplay of exploitation and exploration is referred to as ambidextrous BPM (Grisold et al. 2019, 2022).

With the rapid development of new technologies and digitalization, the BPM discipline is becoming increasingly relevant and therefore must evolve as well (Mending et al. 2020). Especially, new approaches and methods are needed that consider specific purposes/contexts and provide clear practical guidance (vom Brocke et al.

2016, 2021). For a long time, BPM has been criticized for offering rather generic tools with lacking guidance for practical application, although inappropriate BPM methods/tools can lead to wasting organizational resources and even project failure (Schmidt et al. 2001; vom Brocke et al. 2021). It was claimed that one-fits-all approaches do not fulfill the needs of all contexts, particularly not in the digital age (vom Brocke et al. 2016). The shifting purpose in BPM projects and initiatives, from exploitation to exploration, requires the development of novel BPM methods that provide clear guidance while taking into account the situational and organizational context (vom Brocke et al. 2021). Since BPM has traditionally focused on exploiting improvements in existing processes, there is a particular lack of exploratory methods for assessing new technologies' potentials (Grisold et al. 2019; Schmiedel and vom Brocke 2015). The evaluation of 25 BPM methods by Denner et al. (2018b) confirmed that BPM methods are too generic respectively provide little guidance due to lacking applications in specific contexts. Most methods are designed for exploitation and only few for exploration or a joint exploration/exploitation. While most identified methods are moderately context-aware regarding core and management processes, there is no method for support processes. These findings were also confirmed by the extensive analysis of vom Brocke et al. (2021) who reviewed 105 BPM methods as part of their development of a decision support tool, the *Context-Aware BPM Method Assessment and Selection* (CAMAS) method. Apart from classifying the methods based on the context dimensions, the authors also assigned them to the BPM lifecycle stages. The analysis of these methods showed that only few methods were developed for the lifecycle stage of improvement and innovation.

Although literature offers several improvement approaches (Vanwersch et al. 2016; Vergidis et al. 2008), those provide little explicit insights into how improvements or idea generation are actually performed (Vanwersch et al. 2016; Zellner 2011). Due to technological advances, the need for novel methods has dramatically increased in recent years. For instance, researchers proposed methods that investigate the digitization potential of processes or facilitate the detection of use cases for RPA software (Denner et al. 2018a; Herm et al. 2023; Wellmann et al. 2020). However, to the best of our knowledge, there exists no method for embedded analytics. This need is also reflected in the call for criteria for identifying the best processes to automate, organizational scaling approaches, and „[...] critical practices required by digital leaders to ensure strategic automation provides disproportionate business value“ (Lacity & Willcocks 2021, p. 11).

3 Methodology

3.1 Action design research

We followed the action design research (ADR) paradigm and applied the ADR principles introduced by Sein et al. (2011). ADR addresses important real-world problems and aims to develop innovative artifacts extending the theoretical knowledge base related to an abstract class of problems. It is characterized by strong engagement with practitioners and end-users. At the core of ADR studies are the building

of the artefact, the organizational intervention, and its evaluation. Our artifact is a method supporting the identification and selection of business processes that are suitable for embedded analytics and show great improvement potential.

Our ADR study is integrated with an in-depth single case study focusing on the ‘credit and collections’ (C&C) operations at the Hilti Group (Yin 2018). The research project lasted nine months from February until November 2020. It was part of an extended research program conducted by the Hilti Lab for Integrated Performance Management at the University of St. Gallen. The ADR team consisted of the author as the lead researcher, the director of the Hilti Lab, and the group head of controlling of the Hilti Group. Two market organizations (MOs) of Hilti were involved supporting our ADR study as end-users (MO1, MO2). This allowed us to receive input from end-users for building the method artifact as well as to use and evaluate it in real-world interventions. Our research process followed the four ADR stages: 1. *Problem formulation*; 2. *Building, intervention, and evaluation*; 3. *Reflection and learning*; 4. *Formalization of learning* (Sein et al. 2011).

In the first ADR stage, we formulated the problem addressed. In accordance with the ADR principle of practice-inspired research, we highlighted the need for a method that supports organizations in selecting processes for reinvention projects and designing next-generation processes with embedded real-time analytics. We further validated the research opportunity and our research question in a literature review. Besides prior research calls and empirical studies, we validated the need for such a method with representatives of our case company. Following the ADR principle of theory-ingrained artifacts, we further identified existing descriptive and prescriptive knowledge informing the artifact design.

In the second ADR stage, we developed the method artifact. This stage comprised two design cycles that integrated the activities of building, intervention, and evaluation (BIE). We adopted the IT-dominant BIE schema as our research aimed at innovating a BPM method artifact (Sein et al. 2011). While building and intervening were consecutive activities, the evaluation activities were conducted on an ongoing basis. Using situational method engineering (SME), the ADR team developed the alpha version of the method artifact in the first BIE cycle (Henderson-Sellers & Ralyté, 2010). We used this alpha version in a real-world intervention with MO1. The concurrent evaluation activities in design cycle one were formative in nature. First, the method was continuously reviewed by the ADR team while shaping and refining it. Second, we collected feedback from the end-users during and after the intervention with MO1. Overall, the method provided valid and useful results, meaning that the defined method goal was fulfilled. However, further reflection and discussion revealed shortcomings as well. For instance, the end-users criticized that applying the method is quite time-consuming. In particular, we found that a lower level of detail is needed when identifying and documenting processes. This allowed us to significantly accelerate method execution. Furthermore, we removed irrelevant or ambiguous attributes/criteria used in the process assessment model. In consequence, the ADR team decided to execute another BIE cycle. In the second cycle, the ADR team refined the alpha version of the method artifact based on the findings and the feedback gathered in the first BIE cycle. We carried out summative evaluations of the beta version produced. The ADR team performed concurrent

evaluation to determine the method's utility, i.e., the clarity for end-users and the efforts for executing the method. In addition, the ADR team discussed the method's added value with end-users. Then, we applied the refined method in a real-world intervention with MO2. Since no further adjustments were required, the beta version represents our final method artifact and we stopped the second ADR stage after BIE cycle two. By collaborating closely with practitioners and end-users, we fulfilled the ADR principles of reciprocal shaping and mutually influential roles. With the different evaluation steps throughout the two-cycle BIE stage, we also fulfilled the ADR principle of authentic and concurrent evaluation.

The third ADR stage (reflection and learning) paralleled the first two ADR stages. We continuously reflected on the design and redesign of the method and analyzed the intervention results against its goal. As the method design was shaped in an organizational context and respects the feedback from practitioners and end-users, we fulfilled the ADR principle of guided emergence.

In the final ADR stage, we formalized our learning and generalized the research outcomes. Since the individual project characteristics and the organizational context shape situations, we highlight the context-sensitive method elements. In addition, we integrate rich insights from our case study applications into the presentation of the method artifact. We also specify the method's organizational application area and discuss the limitations of our research as well as future research needs. The form and level of detail of the data collected varied across the ADR phases and activities. The building and evaluation activities involved numerous workshops, walkthroughs, and discussions. During the real interventions with the end-users, we collected more structured data via questionnaires, which served as input for our process assessment. Further details on the data gathered from workshops, semi-structured interviews, and questionnaires are provided with the presentation of the method application in Sect. 5.

3.2 Situational method engineering

In general, the term 'method' refers to a series of steps that are taken to achieve a set goal (Braun et al. 2005). A method should have four attributes (i.e., *goal orientation, systematic approach, principles orientation, repeatability*), whereas its design consists of four elements: *activity, tool, role, defined output* (Braun et al. 2005). We adopted SME for building the theory-ingrained method artifact as our research goal was to create a context-sensitive BPM method with a clear purpose (Braun et al. 2005; Bucher et al. 2007; Henderson-Sellers & Ralyté, 2010). Specifically, we carried out method composition, so we selected and composed method fragments matching our situational needs from existing methods (Bucher et al. 2007). Following the assembly-based approach, we built the method artifact in three steps, i.e., specifying the method needs, selecting the method fragments, and assembling the fragments (Henderson-Sellers & Ralyté, 2010).

In an initial step, the ADR team structured the procedural method artifact at an abstract level adapting the BPM lifecycle model by Dumas et al. (2018). For the first two method activities (process identification and process discovery), we

used existing methods from BPM and IS literature. For the third method activity (process analysis), we constructed a novel process assessment model using tools from operational research and management science. The process assessment model is our method's central component and represents the main contribution of this research. First, the ADR team elaborated the method needs and evaluation logic in a causal map (Montibeller & Belton 2006). Then, appropriate process assessment approaches and measures were selected from existing business process improvement methods. We relied on prior literature reviews to identify suitable fragments, i.e., process assessment criteria, and discussed them with practitioners. Furthermore, we designed a new criterion from scratch. The justificatory knowledge informing the method artifact's design is specified with our method presentation in Sect. 4.

3.3 Case study

The Hilti Group is a globally operating company based in Liechtenstein, which offers a wide range of products and services to the construction industry. Hilti employs about 30,000 people and generated net sales of around CHF 6.3 billion in 2022. The project was mandated for a strategic initiative of the leadership team's digital finance agenda at Hilti. Our research addressed the strategic goal of operational excellence leveraging embedded analytics and state-of-the-art technology like ML. The C&C operations were focused for various reasons. The C&C function is extremely important for industrial companies like Hilti since long payment terms are typical in this sector and constitute a competitive advantage. This is also reflected in the high receivables volume of about CHF 2.29 billion, which accounted for almost 34% of the balance sheet total as of December 31, 2022 (according to Hilti Corporation's financial report 2022). In addition, the C&C operations represent the most resource-intensive finance processes. This is also caused by its decentralized organizational structure, whereas most other finance processes were centralized in shared services centers in last decades. Overall, the finance management team believes that the credit processes as well as the collections processes offer substantial improvement opportunities and value potential. Our embedded case study involved interventions with two MOs (MO1, MO2). MO1 and MO2 are mature markets with significant sales volumes and hence large C&C units. In both countries, a senior manager (Head of Credit and Collections) is responsible for these departments that consist of two teams, namely a credit team and a collections team. However, there are relevant contextual differences in terms of cultural and economic conditions. While the economic environment of MO1 is excellent, MO2 operates in a country characterized by weaker macro data. In addition to the weaker economy, differences in culture are also present, e.g., regarding the payment behavior. At MO2, the average days sales outstanding (DSO) reached 84 days, with regular payment terms of 60 days; at MO1, the DSO was approx. 24 days, while payment terms of 30 days are usually granted.

4 Method artifact

4.1 Method overview

Our BPM method aims to select business processes for an improvement/reengineering project, which match with the application scope of embedded analytics and hold high improvement potential as well. Hence, it is a process selection method that guides organizations in their digital transformation journeys, particularly in their initial step of exploring emerging technological opportunities and selecting use cases. The method artifact is a procedural model describing the decision-making process as a sequence of activities. It is split into two decision stages. In decision stage one, potential decision options (processes) are identified and prioritized. In decision stage two, the decision options with high priority are further evaluated. Our method follows the first three stages of the BPM lifecycle model introduced by Dumas et al. (2018), i.e., *process identification*, *process discovery*, and *process analysis*. However, we did not follow the definitions and activities proposed by the authors. Instead, we adapted the activities to our needs. While we used existing knowledge and tools for the first two activities (decision stage one), we built a novel process assessment model for decision stage two. For method engineering, we adopted existing method chunks from theory as well as created a new fragment from scratch.

Table 1 provides an overview of the BPM method artifact. The two decision-making stages are formalized and detailed in the following. We recommend careful planning and scoping when applying the method. Although we provide recommendations on the activities and involved participants, we consider our method to be adaptable. Accordingly, BPM consultants and project manager should consider the individual situational and organizational context and make any changes where necessary. For instance, our case study focuses on two closely related finance functions, the credit management and the collections management at Hilti. We also applied the method with two local market organizations, which allowed us to benchmark them against each other.

4.2 Stage 1: process identification and prioritization

In the first step, the decision options, i.e., the business processes, are identified. If the process architecture is already captured and documented, this step can be skipped. Identifying the current process landscape is carried out during workshops and interviews with the responsible operational managers and process experts. A process classification framework can be a helpful tool in this step. For our case study, we applied the APQC process classification framework (APQC 2018). It includes a list of processes that companies typically employ across industries. It allows a two-way identification ensuring high validity and consistency. On the one hand, the as-is processes can be identified and matched with the processes proposed by the APQC process framework. On the other hand, the generic process list of the framework allows to detect any missed or incomplete processes. The generic processes are just

Table 1 Overview of process selection method

Stage	Activity	Tool	Role	Output
Decision Stage 1 Identification and prioritization of decision options	<i>Process identification</i> Identify processes performed within process groups	Process classification framework	<i>Operational unit:</i> Managers Process experts	<i>Process landscape:</i> Bundle of processes represents total set of decision options
	Map processes in process landscape			
	Prepare and align brief process profiles			
	<i>Process discovery</i> Evaluate processes based on three criteria, i.e., importance, improvement potential, and feasibility			
Decision Stage 2 Evaluation of decision options	Determine the thresholds that must be reached to qualify for assessment	MCDCA ranking method	<i>Management:</i> Senior managers	<i>Process shortlist:</i> Based on high-level evaluation, most relevant/promising processes are selected for evaluation of decision options in stage two
	<i>Process analysis</i> Assessment of shortlisted processes based on criteria that are aggregated to attributes and performance indicators	Process assessment model	<i>Operational unit:</i> Managers Process experts	<i>Process assessment:</i> Results indicate embedded analytics suitability and value potential, supporting profound decision-making
	Analyze single processes and compare decision options			

recommendations and labels or descriptions should be modified where necessary. This ensures the business processes to be clearly delineated and that a common understanding exists among project participants. In this stage, end-users should create process profiles comprising the responsibilities, brief process descriptions, and the main tasks involved. However, there is no need for very detailed information.

In the second activity (process discovery), a high-level assessment of the processes identified is performed to identify the most relevant business processes and to reduce the number of decision options. Thereby, the efforts for executing the third activity (process analysis) in decision stage two are kept within reasonable limits. We adopted the process prioritization and selection method from Dumas et al. (2018), but they intended it for use in the first BPM lifecycle stage (process identification). The authors propose to assess the identified processes based on three generic criteria, i.e., *strategic importance*, *health*, and *feasibility*. The assessment is made by the responsible senior or top managers using semi-structured interviews or questionnaires (on a scale from 0 to 100%). Using these criteria, the set of processes can be ranked/prioritized and plotted in a process portfolio matrix. Furthermore, thresholds can be determined to select the most relevant processes to be assessed in decision stage two.

The criterion *strategic importance* captures the process's relevance for achieving strategic goals and its influence on decisive organizational factors, such as the customer satisfaction or profitability. The criterion *health* provides an indication of the overall process state and performance. It aims to identify processes with critical performance problems and correspondingly high room for improvement. In line with our method objective, we hence relabeled the criterion to *improvement potential*. Finally, the *feasibility* of achieving this improvement potential is evaluated. It depends on a variety of factors. For instance, culture and politics may be crucial obstacles.

4.3 Stage 2: process assessment model

In decision stage two, the pre-selected processes are evaluated in more detail using built process assessment model. In accordance with the purpose of our process selection method, the model addresses the following fundamental question: “*Is this process suitable for using embedded real-time analytics and what is the potential value of a process improvement/reengineering project?*”. The decision problem does not require to select exclusively one process. Instead, decision makers face a set of options (processes) that must be evaluated and compared with each other. Accordingly, the processes selected in decision stage one are assessed regarding their suitability for applying embedded analytics and the accompanying value potential. The data is collected through interviews and questionnaires involving the end-users, i.e., the operational managers and process experts in charge.

In an initial step, we explored the problem structure and elaborated the underlying valuation logic. The literature offers a wide range of methods for structuring problems and evaluating decision options (Marttunen et al. 2017). Causal maps, also known as cognitive maps, are a common method applied to complex decision

problems in multiple disciplines (Montibeller & Belton 2006). They are particularly useful for the initial problem structuring as well as for identifying attributes or criteria that characterize respective decision options. We took a ‘*Value-Focused-Thinking*’ perspective and structured the decision problem underlying our method drawing a causal map in form of a value tree map (Marttunen et al. 2017). The evaluation logic and its causal inferences is specified on three levels (Montibeller & Belton 2006). On the lower level, the attribute concepts are defined (means). These attributes are captured through performance indicators (PIs) and various combined attributes/criteria, respectively. The consequences arising from the attributes are mapped on the second level and aggregated to the overall value potential of a decision option on the third level (end). The causal map was not only an important instrument for building the method but is also an integral conceptual element of the process assessment model proposed. Since not all PIs can be directly measured, we partly built constructs (latent variables) that combine different measures (attributes/criteria) using the value tree map and partly path diagrams (DeVellis 2012). Our assembly-based SME approach involved mainly selecting and composing fragments, i.e., performance indicators and attributes/criteria, from existing methods described in IS and BPM literature (Henderson-Sellers & Ralyté, 2010). The individual method fragments were identified by the ADR research team and selected in collaboration with practitioners and end-users. In addition, we built a novel construct to evaluate the suitability of a process for embedded analytics (PI2) from a behavioral perspective.

The value tree map (Fig. 1) illustrates the following valuation logic. The overall potential value proposition (value concept) is influenced by three performance dimensions (consequence concepts), i.e., *strategic relevance*, *process performance gap*, and *embedded analytics suitability*. To capture these performance dimensions, we specified six PIs (attribute concepts). The value proposition is determined by the process’s *strategic relevance* (PI1) and its improvement potential, referred to as process performance gap. The process performance gap is measured in terms of process *efficiency* (PI2), *quality* (PI3), and *flexibility* (PI4). The applicability of embedded real-time analytics (embedded analytics suitability) is assessed considering the *technical applicability* (PI5) and the *task-analytics-fit* (PI6).

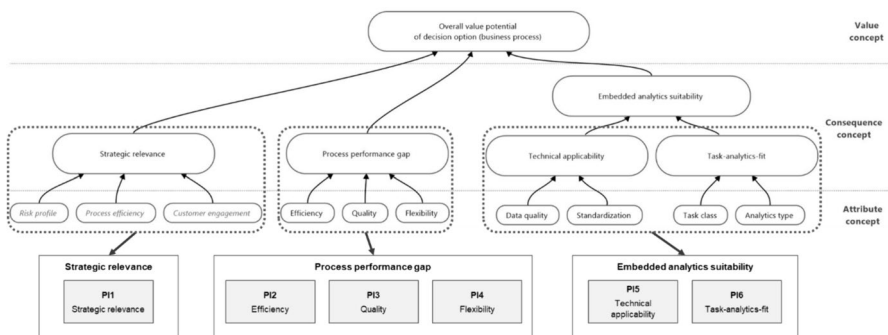


Fig. 1 Value tree map and performance indicators

The *strategic relevance* has decisive impact on the overall value proposition. It works like a multiplier—the higher the strategic relevance, the greater the value proposition from process improvement projects. Literature provides several approaches and measures used to determine this factor. We follow the definition of Zelt et al. (2018), who define process importance as one out of five generic process dimensions. Accordingly, strategic relevance reflects the process's impact on organizational competitiveness and combines the aspects of criticality and value contribution. In the ADR design cycles with practitioners and end-users, we found that this performance dimension is highly context-sensitive and depends on the individual performance management system in place. Consequently, it should be specified depending on the individual BPM context (vom Brocke et al. 2016, 2021). In our case study, we determined the strategic relevance using the internal target system of our end users. This is formed by the corporate strategy, the organizational performance management system, and the application domain focused.

To assess the *process performance gap*, we rely on BPM fundamental knowledge. Performance measurement is a crucial subject in the BPM discipline and the literature offers a wide range of generic as well as context-specific process performance measures (Dumas et al. 2018; Zelt et al. 2018). Dumas et al. (2018) list four generic process performance dimensions (time, cost, quality, and flexibility), which can be assessed in an aggregated manner as well as with specified performance measures. These process performance dimensions informed the PIs 2, 3, and 4. We applied these generic PIs to ensure that our method is also suitable in other contexts and to provide room for adaptations. We combine time and costs in PI2 (efficiency) to capture the process performance holistically while taking individual aspects into account. Quality (PI3) can be assessed from the various perspectives. In particular, the process participants and customer perspective must be respected (Dumas et al. 2018). Flexibility can generally be described as the ability to respond to internal and external changes. In the context of our process assessment model, PI4 (flexibility) assesses the ability to handle different cases and fluctuating workloads as well as the ability of managers to change structures and allocation rules. However, also other types of flexibility may be considered. For each PI, we assess the 'actual condition (as-is)' and the estimated 'improvement potential'. We chose these criteria as they allow for an aggregated and consistent assessment. By contrast, the actual process performance measures used in our case company were not suitable as they are highly specific and correspondingly less generic.

The performance dimension *embedded analytics suitability* addresses our method's exploratory purpose examining if, and to what extent, embedded analytics is applicable. The automation literature provides various methods and criteria for the identification and selection of RPA use cases (Fung 2014; Herm et al. 2023; Leshob et al. 2018; Santos et al. 2020). Apart from these dedicated methods, further IS research informed the design of both PIs. Technical applicability (PI5) is evaluated in terms of data quality and process standardization. We found that data quality is a major factor for applying embedded analytics. Depending on the individual context, the term data quality can have varying meanings (Strong et al. 1997). In our process selection method, data quality refers to the prerequisite that a digital database exists. For evaluation, we apply a liked scale that maps the extent of analog

data and digital data used in a process (Cappiello et al. 2004). Furthermore, technical applicability is assessed from a task perspective. Despite varying definitions and evaluation approaches, there is a consensus that the complexity of the tasks involved in a process shapes its standardization (Liu & Li 2012). We adopt the interpretation of process standardization from automation literature (Lacity & Willcocks 2016; Santos et al. 2020; Wellmann et al. 2020). The degree of process standardization is expressed by a liked scale, which allows standardized processes (include only simple tasks) and complex processes (include cognitive tasks) to be differentiated.

Although process standardization is a suitable criterion for identifying simple processes and RPA use cases, it is not clear whether analytics is applicable to complex tasks involving cognitive intelligence. To the best of our knowledge, literature lacks a measure or evaluation logic for assessing the fit of analytics applications to different task classes. Therefore, we innovated a novel construct, referred to as ‘task-analytics-fit’ (PI6). It is inspired by the ‘task-technology-fit’ concept that is long-established in IS research (Goodhue & Thompson 1995). Combining existing theoretical concepts, we built a linking model that maps task classes with different analytics application types (Fig. 2). From an information-processing perspective, tasks can be categorized by their complexity (Campbell 1988; Liu & Li 2012). It allows simple tasks with deterministic outcomes to be differentiated from complex tasks with probabilistic outcomes (Lacity & Willcocks 2016). First, we categorize the main tasks involved in the processes prioritized in decision stage one, in order to separate simple from complex tasks. The process profiles prepared can support this step. Then, the complex tasks identified are further classified. We adapted the model proposed by Ham et al. (2012) and recognize four cognitive information processing activities, referred to as task classes: I. Information processing and analysis (descriptive); II. Information analysis (diagnostic); III. judgment and decision/

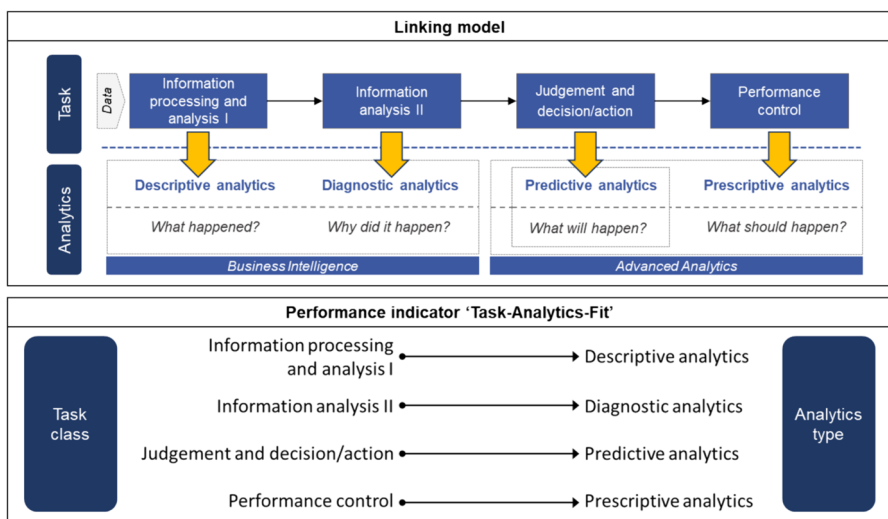


Fig. 2 Linking model and performance indicator ‘task-analytics-fit’

action; IV. Performance control. Furthermore, we adopted the business analytics application types proposed in literature, i.e., descriptive, diagnostic, predictive, and prescriptive analytics (Banerjee et al. 2013; Delen & Demirkan 2013). Our linking model connects the task classes with the analytics application types. In this way, it is possible to determine potentially appropriate analytics types for the complex tasks involved in a process. Fit is hence understood as the matching of two theoretically related variables (Zigurs & Buckland 1998).

Table 2 provides an overview of our process assessment model with the PIs and attributes/criteria as well as the evaluation logics and scales used. Where PIs or individual attribute constructs were not directly measured, we aggregated the PIs/attributes by averaging the underlying attributes/criteria. This approach was also applied when data was collected from multiple project participants. We are aware of the associated shortcomings, however, the ADR research team and case study participants considered this approach to be appropriate for our purpose. Nevertheless, we emphasize that the PIs respectively the attributes/criteria, along with the evaluation approaches and scales used, should be validated and adapted to the individual project and organizational context (Bucher et al. 2007; vom Brocke et al. 2021).

5 Case study: credit and collections operations at Hilti

5.1 Stage 1: process identification and prioritization

ADR is characterized by intense interaction and collaboration with practitioners and end-users. The ADR research team iteratively developed the method artifact in an in-depth case study with the Hilti Group. This involved real-world interventions to use and evaluate the artifact in a naturalistic setting with end-users. In the following, we illustrate our end-user interventions with two local Hilti entities (MO1, MO2). In Sect. 5.1 and Sect. 5.2, we present the method execution and outcomes of our real-world interventions with MO1. In Sect. 5.3, we perform a comparative analysis of both intervention outcomes and draw conclusions for the Hilti's C&C operations on an organizational level.

In the first method activity, we used the APQC classification framework to build up the C&C process landscapes of involved Hilti subsidiaries (APQC 2018). This step was performed with local C&C managers and process experts. The C&C processes are assigned to the APQC process category '9.0 Manage financial resources' (APQC 2019). We adopted the processes proposed, however, we modified some generic process labels and specified the process descriptions to our individual case study context. The processes were identified and recorded in the course of workshops, focus group discussions, and document analysis. The process lists contain process profiles with brief descriptions of their function and main tasks. It must be stated that the process identification was the most time-consuming method activity. However, the effort declined in the second intervention with MO2 as we benefitted from our experience with MO1.

Figure 3 illustrates MO1's C&C process landscape with the 17 C&C-related processes identified. These processes represent the full set of decision options, which

Table 2 Process assessment model

Performance dimension	Performance indicator	Attribute/criteria	Evaluation logic	Scale
Strategic relevance	PI1 Strategic relevance	Strategic relevance (<i>specified in case study context</i>)	low vs. high	1 to 5
	PI2 Efficiency	Actual condition (as-is) Improvement potential	dissatisfying vs. satisfying low vs. high	1 to 5 -2 to +2
Process performance gap	PI3 Quality	Actual condition (as-is) Improvement potential	dissatisfying vs. satisfying low vs. high	1 to 5 -2 to +2
	PI4 Flexibility	Actual condition (as-is) Improvement potential	dissatisfying vs. satisfying low vs. high	1 to 5 -2 to +2
	PI5 Technical applicability	Data quality Process standardization	analog data vs. digital data Simple vs. complex	1 to 5 1 to 5
	PI6 Task-analytics-fit	<i>Task class:</i> Information processing and analysis I Information analysis II Judgement and decision/action Performance control	<i>Analytics types:</i> Descriptive analytics Diagnostic analytics Predictive analytics Prescriptive analytics	Linking model

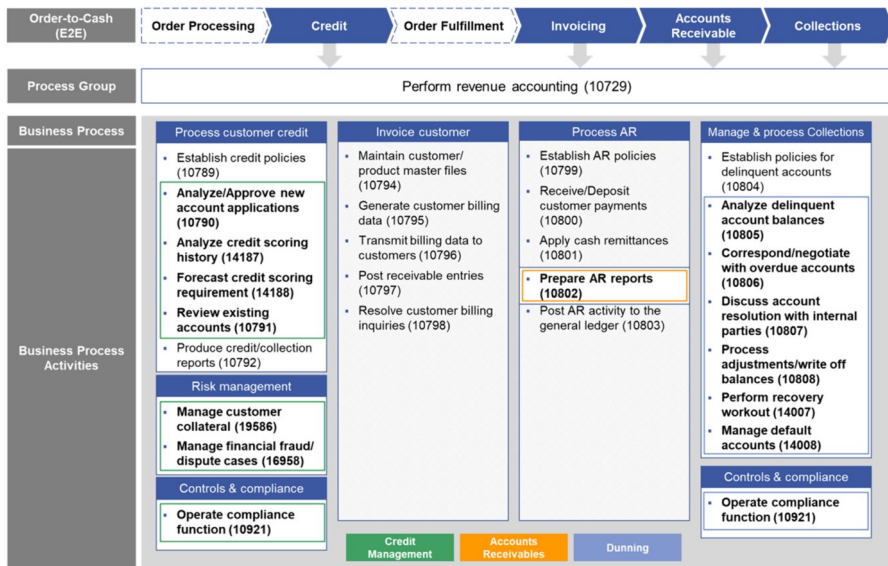


Fig. 3 C&C process landscape

were evaluated in method activity two to select the most critical ones for the detailed analysis in decision stage two. In this step, the functional senior management (local C&C heads), the regional CFO, and the Group Head of Controlling evaluated the identified processes against three criteria: importance, improvement potential, and feasibility. We used a scale of 0–100% and averaged the individual evaluations. In addition, the regional CFOs set thresholds for these criteria, which were used for filtering the list of processes. In total, nine processes missed these thresholds, thus only eight decision options remained. Consequently, the effort required for evaluating relevant decision options in decision stage two was reduced and limited using this simple ranking method. Figure 4 plots the criteria assessed in shape of a process portfolio matrix and lists the eight processes exceeding the criteria thresholds set. It became clear that the collections processes, and especially the ‘dunning process’



Fig. 4 Process portfolio matrix and process shortlist

(1), are highly important. The improvement potential is moderate but feasible. With regards to the credit processes, it is remarkable that the processes of ‘credit risk analysis’ (4) and ‘credit decision-making’ (5) offer high improvement potential, however, their feasibility was less optimistically assessed by the interviewees.

5.2 Stage 2: process analysis

In decision stage two, the eight processes prioritized are analyzed in more detail. Using the process assessment model proposed, the processes are evaluated in terms of strategic relevance, process performance gap, and embedded analytics suitability. This step was performed with local C&C managers and process experts. This step was performed with local C&C managers and process experts. The input data were collected by questionnaires. For each process, we assessed the PIs based on surveyed attributes/criteria. If needed, we aggregated the PIs/attributes by averaging the underlying attributes/criteria. This allowed us to capture the problem in its breadth while making the identified processes comparable. Despite their indeterminacy, PIs are well suited to our complex decision problem as the decision options do not directly compete with each others. The first performance dimension is concerned with the strategic relevance, which we measured based on the target system in place. We adopted the three C&C target system’s objectives as assessment criteria, i.e. risk profile, efficiency, and customer engagement. The process performance gap, or improvement potential, comprised three PIs (efficiency, quality, and flexibility). The embedded analytics suitability was measured using two PIs, assessing the technical applicability and the task-analytics-fit.

For MO1, the results of the third method activity (process analysis) are as follows. Four credit processes were evaluated, namely ‘analyze and approve credit applications by new accounts’, ‘credit risk assessment’, ‘credit decision-making’, and ‘credit risk monitoring’. The credit processes have a high overall strategic relevance. They are characterized by an overall good efficiency thanks to a high degree of standardization and automation. There are a few systemic weaknesses that cause extra manual work, e.g., due to missing interfaces between the credit management software and internal/external information sources. Quality accounts for the greatest portion of the process performance gap, whereby improvement opportunities relate to both the professionalization of the credit management function and the flexibility in managing operations and processes. However, any actions taken to improve quality must ensure that efficiency and integration with existing credit operations and processes are given. We evaluated three collections processes, i.e., ‘analyze overdue account balances’, ‘dunning process’, and ‘adjustments/write-off balance’. The results produced are similar to the credit processes. The collections processes are predominantly of high strategic relevance, with the dunning process being particularly important due its direct customer interaction. The operational process performance gap is moderate. Most collections processes show an adequate automation level, which is partly at the expense of process quality. For instance, the dunning process’s simplified process logic lowers the effectiveness of performed dunning activities. In addition, the collections processes’ value potential is driven by their

strategic importance. All in all, the credit processes are highly suitable for embedded analytics. Credit processes by nature involve various analytical tasks requiring cognitive intelligence. Accordingly, they are considered to be highly suitable for embedded analytics. BI is applicable to various tasks in the main credit processes. Moreover, predictive analytics and prescriptive analytics offer considerable optimization potential for the processes of ‘credit risk analysis’ and ‘credit decision-making’. The collections processes are highly suitable for embedded analytics as well. BI is particularly appropriate for analyzing overdue receivables on both the customer and the portfolio level. In addition, predictive analytics and prescriptive analytics provide significant potential for improving the dunning process. Overall, embedded analytics offers various opportunities for improving the collections management function. Figure 5 illustrates the process assessment for the dunning process at MO1.

5.3 Benchmarking and process selection

Following the first real-world intervention with MO1, the ADR research team refined the method artifact in the second design cycle. It involved another intervention with end-users (MO2), which enabled us to benchmark both C&C units. The comparative analysis enabled us to identify differences and similarities in the C&C process landscapes of both subsidiaries, thereby widening the decision-making information base. Since MO1 and MO2 represent large Hilti markets with different cultural and economic contexts, the method outcomes also provide an indication regarding the entire C&C operations at Hilti and the overall value potential of individual decision options.

In decision stage one, the method produced similar outcomes for MO1 and MO2. The processes identified were mostly identical, thus the C&C process landscapes of both Hilti subsidiaries just slightly differ. Similarly, the high-level evaluation by the C&C management, the regional CFO, and the Group Head of Controlling yielded

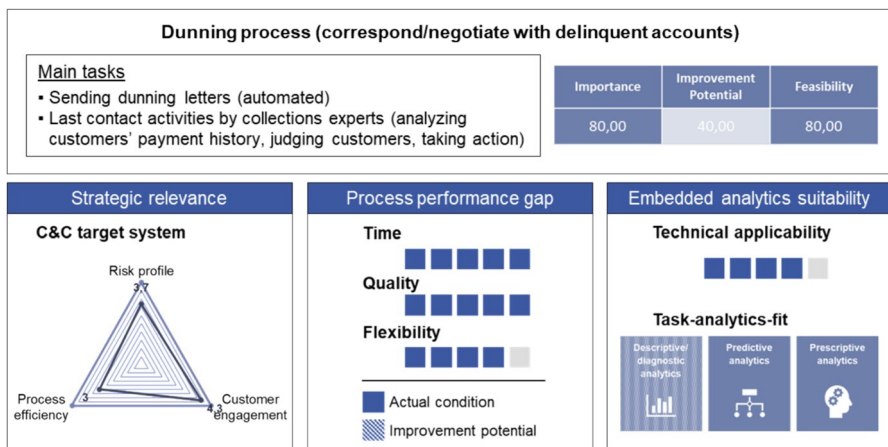


Fig. 5 Exemplary process assessment

also quite consistent results and the processes prioritized were almost identical. The same is true for the results of our process analysis conducted in decision stage two. On the one hand, the assessments of the prioritized processes' strategic relevance yielded quite consistent results. On the other hand, the process performance gaps were similarly assessed in both cases. It can hence be concluded that there is a relatively high level of process harmonization between MO1 and MO2. Nonetheless, fundamental differences caused by divergent cultural and economic contexts became evident. The improvement potential hence varies as MO1 and MO2 suffer from differing weaknesses and pain points, respectively. It is less astonishing that our assessment model produced almost identical outcomes concerning the embedded analytics suitability. The task-analytics-fit was found to be virtually the same as complex tasks involved are identical to a large extent.

Based on the comprehensive process analysis conducted with two representative Hilti subsidiaries, the finance leadership team decided to initiate a reengineering project to improve the dunning process. Meanwhile, the conceptualization and prototyping of a smart dunning process leveraging embedded real-time analytics has started. In this project, the initial focus is on MO1 but there are plans to scale up such a solution within the Hilti Group in future.

6 Discussion

This research investigates how organizations can identify and select embedded analytics use cases. In the two-cycle ADR study, we iteratively built, used, evaluated, and refined an exploratory BPM method artifact, namely a process selection method. The theory-ingrained method artifact is a decision support tool to select operational business processes, for which a reengineering project should be initiated. In interventions with two C&C units of local Hilti subsidiaries, the method artifact was used and evaluated in the real world. While we adopted existing methods for the first two activities, process identification and process discovery, we built a novel process assessment model for the third activity (process analysis) composing method chunks and theoretical concepts from IS and BPM literature. The process assessment model is designed to evaluate business processes (decision options) regarding their embedded analytics suitability and inherent value potential. We identified three performance dimensions (strategic relevance, process performance gap, embedded analytics suitability), which are represented by six PIs. Five PIs were constructed by selecting and composing attributes/criteria from existing methods. In contrast, we conceptualized and constructed a novel PI assessing the task-analytics-fit based on a logical model that integrates existing theoretical concepts.

The literature offers several methods for analyzing processes and exploiting improvement potentials (Vanwersch et al. 2016; Zellner 2011). However, in light of rapid technological progress, several scholars called for further research on opportunity-driven BPM methods enabling to explore emerging technologies' potentials (Denner et al. 2018b; Grisold et al. 2019, 2022). For instance, researchers proposed methods for exploiting business processes' digitalization potentials or for identifying RPA use cases (Denner et al. 2018a; Herm et al. 2023; Wellmann et al.

2020). However, methods related to business analytics are lacking so far. This article addresses this gap by introducing a process selection method. We hence contribute an exploratory method artifact to the BPM body of knowledge and, more specifically, to the research field of BPM capabilities and methods. With respect to the hyperautomation phenomenon, we extend the literature discussing methods supporting the identification and implementation of automation use cases. The process assessment model and especially the conceptualized measure of ‘task-analytics-fit’ can be considered as a main innovation.

Our research has important practical implications as well. The process selection method provides organizations with important guidance in their digital transformation journey, especially in the initial adoption stage. By identifying, prioritizing, and analyzing several decision options, it provides a comprehensive information base for decision-making. In addition, the process assessment model and the ‘task-analytics-fit’ construct could be applied on a stand-alone basis, e.g., to conduct ad-hoc process assessments.

These theoretical and practical implications are contrasted by various limitations. The limitations result from our individual case study context and particularly the process context. Our case study focused on the C&C operations at the Hilti Group. The C&C processes represent transactional support processes involving several complex tasks requiring cognitive intelligence and expert knowledge. Hence, C&C processes are well suited for analytics applications by nature. In contrast, we expect our method to be less suitable for processes that, for example, involve emotional or social intelligence (Kaplan & Haenlein 2019). We also see limitations related to the process assessment model. Although we consider the novel PI (task-analytics-fit) to be generic, the other PIs are highly context-dependent and less robust. Correspondingly, we see two directions for future research. First, our process selection method should be applied in other companies and contexts to validate its generalizability and further develop the method. Second, we would appreciate further research focusing on the process assessment model proposed. Especially, validating and refining the novel construct (task-analytics-fit) would be of great relevance for both academics and practitioners.

7 Conclusion

This article introduces an exploratory BPM method, i.e., a process selection method for embedded analytics. It provides organizations with guidance in identifying and selecting operational business processes, for which a reengineering project should be initiated. Applying ADR and SME, we iteratively built, used, evaluated, and refined the theory-ingrained method artifact. At its core is a process assessment model that we composed based on existing method chunks and a novel construct evaluating the ‘task-analytics-fit’. In two interventions with local Hilti subsidiaries, we used and evaluated the process selection method in the real world. Our research hence contributes an exploratory method that extends the BPM knowledge base.

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Data availability The data generated and analyzed during the study are not publicly available due to confidential company data but are provided by the corresponding author upon request.

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